AICTE IBM SkillsBuild Internship

PROBLEM STATEMENT & SOLUTION PROPOSAL

ContractIQ Sentinel

AI-Powered Enterprise Contract Intelligence Platform

Project Domain

Enterprise AI | Legal Technology | Document Intelligence

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1. Introduction

Contracts govern nearly every material relationship an enterprise enters into — with vendors, employees, customers, and regulators. A single overlooked clause in a procurement agreement or a missed compliance obligation in a vendor contract can expose an organization to financial penalties, legal disputes, or reputational damage. As enterprises scale, the volume of contracts they generate and receive grows far faster than the legal and procurement teams available to review them.

Manual contract review remains the industry norm across most mid-size and large organizations. Legal teams read agreements clause by clause, cross-reference them against internal policy templates, and manually flag deviations. This approach is reliable when performed carefully, but it does not scale: reviewers face fatigue over long documents, interpretation varies from one reviewer to another, and turnaround time increases directly with contract volume and complexity.

The core challenges of manual review can be summarized as follows:

- **Inconsistent interpretation** — different reviewers apply different judgment to the same clause type.
- **Slow turnaround** — high-value deals are delayed while legal teams work through a backlog.
- **Limited visibility** — leadership rarely has an aggregated, real-time view of contractual risk across the organization.
- **Human error** — critical clauses such as indemnification, liability caps, or termination rights can be missed under time pressure.

These constraints create a clear need for an AI-powered contract intelligence layer that can read, interpret, and assess contracts with the consistency of a rules engine and the contextual understanding of a language model. **ContractIQ Sentinel** is designed to fill this gap, using IBM's enterprise-grade AI stack to automate the parts of contract review that are repetitive and rules-based, while surfacing the judgment calls that still require human sign-off.

2. Problem Statement

2.1 The Challenge

Organizations that process a high volume of contracts — procurement agreements, vendor MSAs, employment contracts, NDAs — face a recurring set of operational problems that manual review cannot resolve at scale.

Challenge	Impact on the Business
Manual, clause-by-clause review	Review cycles stretch from days to weeks for lengthy agreements.
Human error under time pressure	Missed liability, indemnity, or termination clauses create legal exposure.
Compliance gaps	Deviations from mandatory regulatory or internal policy clauses go undetected.
Clause inconsistency across contracts	No standard baseline to compare a new contract against approved templates.
No aggregated risk visibility	Leadership cannot see contractual risk exposure across the organization in real time.

2.2 The Objective

The objective of ContractIQ Sentinel is to give legal, procurement, and compliance teams an AI-assisted first pass over every contract that enters the organization, so that human reviewers spend their time on judgment calls rather than manual extraction. Specifically, the platform aims to:

- Automatically extract and structure clauses from uploaded contracts in PDF or DOCX format.
- Detect deviations from an organization's standard clause templates.
- Assign a quantified risk score to each contract and to individual clauses.
- Generate an executive summary and actionable recommendations for legal reviewers.
- Produce exportable Excel reports and a live dashboard for portfolio-level visibility.

3. Proposed Solution

ContractIQ Sentinel is architected as a full-stack web application with an agentic AI core built on IBM watsonx Orchestrate and IBM Granite models. The application accepts a contract upload, extracts and parses its content, and routes the parsed clauses through an orchestrated AI workflow that performs deviation detection, risk assessment, and recommendation generation, before returning results to the user through an interactive dashboard and a downloadable Excel report.

End-to-End Workflow

- **Upload:** The user uploads a contract (PDF or DOCX) through the React frontend.
- **Extraction:** The Node.js/Express backend extracts raw text from the document.
- **Orchestration:** IBM watsonx Orchestrate coordinates the downstream AI workflow, sequencing clause parsing, deviation checks, and risk analysis as connected agentic tasks.
- **Language Understanding:** IBM Granite models interpret clause language, identify clause types, and compare them against standard templates.
- **Risk Assessment:** The Contract Intelligence Engine scores each clause and rolls up an overall contract risk rating.
- **Storage:** Parsed clauses, scores, and metadata are persisted in MongoDB Atlas.
- **Reporting:** An Excel report and executive summary are generated and made available on the dashboard.

This workflow keeps the user experience simple — upload a document, receive a structured risk assessment — while the orchestration layer underneath coordinates several discrete AI tasks that would otherwise require separate manual steps.

4. Technology Stack

The platform is built entirely on cloud-native, industry-standard technologies, combined with IBM's enterprise AI services for the intelligence layer.

Technology	Purpose
React	Component-based frontend framework for the user interface.
Vite	Fast build tooling and development server for the React application.
Material UI	Pre-built, accessible UI component library for consistent design.
Node.js	JavaScript runtime powering the backend services.

Technology	Purpose
Express.js	Web framework for building REST APIs on Node.js.
MongoDB Atlas	Managed cloud database for contracts, clauses, and metadata.
IBM watsonx Orchestrate	Agentic orchestration of the contract analysis workflow.
IBM Granite	Foundation models for clause understanding and language tasks.
IBM Bob	AI-assisted development tool used during platform build-out.
Render	Cloud hosting for the backend API.
Vercel	Cloud hosting and CDN for the frontend application.
GitHub	Source control and CI/CD pipeline integration.

5. System Architecture

ContractIQ Sentinel follows a layered architecture: a presentation layer (React), an application layer (Node.js/Express), an agentic AI orchestration layer (IBM watsonx Orchestrate and IBM Granite), a domain-specific processing layer (the Contract Intelligence Engine), and a persistence and reporting layer (MongoDB Atlas, Excel reports, and the dashboard). Each layer communicates over well-defined REST interfaces, which keeps the system modular and allows any single layer to be scaled or replaced independently.

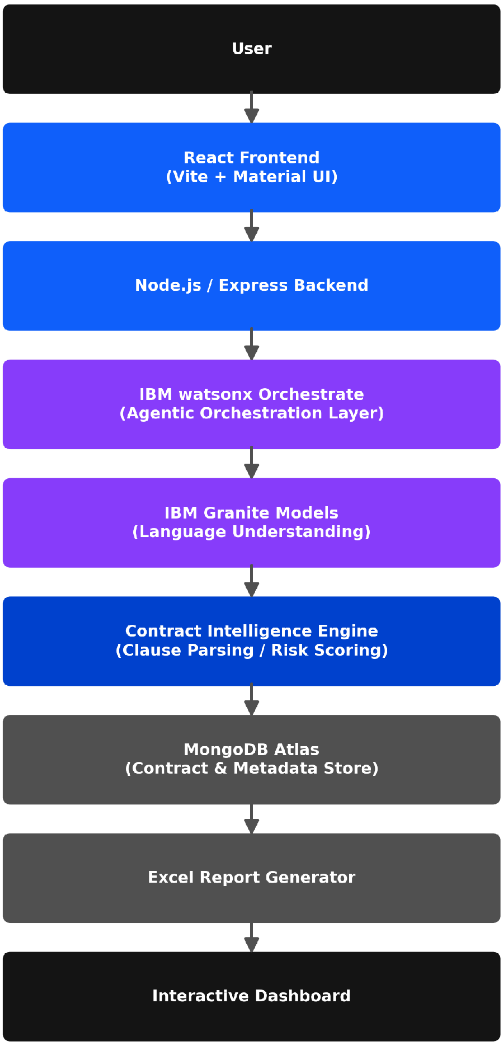

Figure 1: ContractIQ Sentinel high-level system architecture.

6. IBM Components Used

The intelligence layer of ContractIQ Sentinel is built directly on IBM's AI stack. Each component plays a distinct role in the agentic workflow:

Component	Role
Chat Input	Captures the user's request and contract context entering the watsonx Orchestrate workflow.
IBM watsonx AI Agent	Coordinates the sequence of clause extraction, risk assessment, and summary tasks.
IBM Granite Model	Performs the underlying natural-language understanding of clause text.
Chat Output	Returns the structured recommendation and summary back to the application layer.
Knowledge Base	Stores standard clause templates used as the reference baseline for deviation checks.
Excel Report Generator	Converts the structured analysis into a downloadable, formatted Excel report.

7. Role of Agentic AI

IBM watsonx Orchestrate acts as the coordination layer that sequences the individual AI tasks required to move from a raw contract upload to a finished executive summary. Rather than a single monolithic model call, the workflow is broken into discrete agentic steps:

- **Document understanding** — interpreting the structure of the uploaded contract.
- **Clause extraction** — identifying and isolating individual clauses by type.
- **Risk assessment** — scoring each clause against known risk patterns and templates.
- **Executive summary generation** — condensing findings into a business-readable summary.
- **Recommendation generation** — proposing specific redline or negotiation actions.

This is meaningfully different from a traditional document parser, which typically applies fixed regular expressions or keyword rules to extract text. A traditional parser can locate where a clause begins, but it cannot reason about whether the clause's terms are favorable, standard, or high-risk. Agentic orchestration allows each step to use the appropriate model capability — language understanding for clause interpretation, reasoning for risk scoring — and to pass structured context between steps, producing an assessment that reflects contractual judgment rather than pattern matching alone.

8. Novelty and Uniqueness

ContractIQ Sentinel differentiates itself from conventional contract management software through the depth of its AI-driven analysis rather than simple document storage and search:

- **AI-powered contract intelligence** that reads and interprets clause content, not just document metadata.
- **Clause deviation detection** against organization-defined standard templates.
- **Automated risk scoring** at both the clause and contract level.
- **Executive report generation** that translates legal detail into business-readable summaries.

- **An enterprise dashboard** giving portfolio-wide visibility into contractual risk.
- **Cloud-native deployment** across Vercel, Render, and MongoDB Atlas for elastic scaling.
- **Native IBM AI integration** through watsonx Orchestrate and Granite, aligned with enterprise AI governance standards.

9. Scalability

The platform is designed from the outset for enterprise-scale deployment rather than single-user use:

- **Cloud-native architecture** — frontend, backend, and database are independently deployed and independently scalable.
- **Multi-user support** — the backend and database schema are designed to support concurrent users and organizations.
- **Enterprise deployment readiness** — the stack maps directly onto standard enterprise cloud deployment patterns.
- **API extensibility** — the Express REST API can expose endpoints for integration with existing procurement or CLM systems.
- **Modular AI architecture** — the watsonx Orchestrate workflow can add new agentic tasks (for example, a new clause type check) without redesigning the core pipeline.

10. Commercial Viability

Target Customers

- Enterprises with high-volume vendor and procurement contracting.
- Legal firms managing contract review for multiple clients.
- Procurement teams evaluating supplier agreements at scale.
- Government organizations with compliance-heavy contracting requirements.
- Financial institutions subject to strict regulatory contract review.

Business Model

- **SaaS subscription** — tiered monthly or annual pricing based on contract volume.
- **Enterprise licensing** — dedicated deployment and support agreements for large organizations.
- **API subscriptions** — usage-based pricing for organizations integrating ContractIQ Sentinel into existing systems.

11. Contribution to Society

Beyond its commercial application, ContractIQ Sentinel contributes to more reliable and transparent business practices:

- **Reduces legal errors** by catching clause-level risks that manual review under time pressure can miss.
- **Improves regulatory compliance** by systematically checking contracts against required policy clauses.
- **Saves organizational time**, freeing legal teams to focus on negotiation and strategy rather than manual extraction.
- **Supports digital transformation** in legal and procurement functions that have historically remained paper-heavy.

- **Enables better decision-making** by giving leadership a consolidated, evidence-based view of contractual risk.

12. Readiness for Deployment

Layer	Platform
Frontend	Vercel — global CDN hosting for the React application.
Backend	Render — managed hosting for the Node.js/Express API.
Database	MongoDB Atlas — managed, cloud-hosted document database.

Each component of ContractIQ Sentinel is already mapped to a specific managed cloud platform rather than requiring custom infrastructure. Because the frontend, backend, and database are decoupled and independently deployable, the solution can move from a working prototype to a production deployment without architectural rework, making it deployment-ready for a pilot rollout.

13. Future Scope

Planned enhancements beyond the current scope of ContractIQ Sentinel include:

- **OCR support** for scanned and image-based contracts.
- **An AI Legal Copilot** for conversational, in-context contract queries.
- **Fine-tuned Granite models** trained on organization-specific clause libraries.
- **Multi-language contract support** for global enterprise operations.
- **Digital signature integration** for end-to-end contract lifecycle management.
- **Real-time compliance monitoring** as regulations and internal policies change.

14. Conclusion

ContractIQ Sentinel addresses a well-defined, high-cost problem — slow, inconsistent, and error-prone manual contract review — with an architecture that combines standard cloud-native web technologies with IBM's enterprise AI stack. By orchestrating clause extraction, deviation detection, risk scoring, and executive summary generation through IBM watsonx Orchestrate and IBM Granite, the platform reduces the manual burden on legal and procurement teams while giving organizations a consolidated view of contractual risk. The solution is scalable, deployment-ready, and commercially viable across a range of enterprise and institutional customers, making it a strong candidate for continued development beyond the scope of this internship.

15. References

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